Course Outcomes

M. Sc. (Hons) in Computer Science (Specialization in Data Science) under the framework of HonoursSchool System

FIRST SEMESTER

Paper Title: Principles of Data Science

Paper Code: MDS – 2401

Course Outcomes: After successful completion of the course, student will be able to:

- Understand the impact of digital data in businesses its role in making crucial decisions.
- Have preliminary knowledge on data science process, tools, and job trends.
- Have proficiency in utilizing diverse feature selection, anomaly detection methods and timeseries forecasting techniques.
- Have knowledge about available data science tools.

Paper Title: Programming in Python

Paper Code: MDS – 2402

Course Outcomes: After successful completion of the course, student will be able to:

● Demonstrate Python Programming syntax, including variables, data types, andoperators, enabling them to write clear and concise Python code.

● Develop strong problem-solving skills by applying control flow structures andfunctions to design solutions for a variety of computational problems.

● Use fundamental data structures in Python, such as lists, tuples, dictionaries, and sets,and will be able to manipulate and analyze data effectively.

● Acquire a solid foundation in object-oriented programming (OOP), understanding howto create and use classes, objects, inheritance, and polymorphism in Python forbuilding modular and scalable programs.

Paper Title: Advance Database Systems

Paper Code: MDS – 2403

Course Outcomes: After completing this course the students will be able to:

- **Demonstrate proficiency in advanced database design, optimization, and implementation,showcasing their ability to architect robust and scalable data systems.**
- **Apply their knowledge to implement and manage databases using emerging technologies, suchas NoSQL, distributed databases, and in-memory databases.**
- **Exhibit leadership in data security and compliance, implementing advanced measures tosafeguard sensitive information in accordance with industry standards and regulations.**

- **Possess strategic data management skills, ensuring their ability to handle large-scale datasetsefficiently.**

Paper Title: Operating System with Linux

Paper Code: MDS – 2404

Course Outcomes: After successful completion of course the students will be able to:

- **Describe the important computer system resources and the role of operating system in theirmanagement policies and algorithms.**
- **Understand the process management policies and scheduling of processes by CPU**
- **Evaluate the requirement for process synchronization and coordination, memory managementhandled by operating system also cater-policies with respect to different storage.**
- **Learn to use features of Linux for programming and administration**

SECOND SEMESTER

Paper Title: Analysis and Design of Algorithms

Paper Code : MDS – 2408

Course Outcomes : After successful completion of course the students will be able to:

- **Analysis of algorithms including recursive algorithms, space and time complexity.**
- **Understand the Divide and Conquer design strategy to solve different problems like sorting,selection and matrix multiplication.**
- **The students will also be able to learn various Dynamic Programming design techniques.**
- **Backtracking and Branch n Bound design techniques and application of these techniques tosolve problems. Deal with finding lower bounds on complexity for sorting and search problemsusing comparison trees and the lower bounds on parallel computations.**
- **NP-Hard and NP-Complete problems and the examples of such problems able be written.**

Paper Title: Statistical Methods for Data Science

Paper Code : MDS – 2409

Course Outcomes: After successful completion of course the students will be able to:

- **Conceptual understanding of R language and its understanding for statistical computing andgraphics. Implementation of R fundamentals and data frames.**
- **Knowledge of hypothesis testing and curve fitting.**
- **Understanding of important industrial use concepts as regression, classification, dimensionreduction and clustering.**
- **Understanding of important statistical based machine learning concepts as support vectormachines, decision trees and random forest techniques.**

Paper Title: Data Mining and Artificial Intelligence

Paper Code : MDS – 2410

Course outcomes: After completing the course the student will be able to:

- **Have understanding of data warehousing fundamentals and data mining techniques likeassociation, classification, and clustering.**
- **Have knowledge of Artificial intelligence historical context, heuristic search methods, andknowledge representation encompassing logic and semantic networks.**
- **Have possess knowledge of specialized AI areas, including expert systems, natural languageprocessing, genetic algorithms, and artificial neural networks covering various learningparadigms.**

Paper Title: Big Data Analytics

Paper Code : MDS – 2411

Course Outcome: After the completion of the course the students will be able to

- **Understand the key issues in big data management and its associated applications in intelligentbusiness and scientific computing.**
- **Acquire fundamental enabling techniques and scalable algorithms like Hadoop, and MapReduce in big data analytics.**
- **Interpret business models and scientific computing paradigms, and apply software tools for bigdata analytics.**
- **Achieve adequate perspectives of big data analytics in various applications like recommendersystems, social media applications etc.**

THIRD SEMESTER

Paper Title: Data Visualization

Paper Code : MDS - 2415

Course Outcomes: After the completion of the course the students will

- **Have understanding of data visualization techniques to develop charts, maps, plots, and other visual representations of data.**
- **Be able to implement the data visualization techniques using Python libraries and Tableau.**
- **Have knowledge of application of different visualization approaches on real-world datasets.**

Paper Title: Machine Learning: Tools and Techniques

Paper Code: MDS – 2416

Course Outcomes: After completion of the course the student will be able to:

- **Comprehend core machine learning principles spanning supervised, unsupervised, and deep learning techniques.**
- **Apply diverse machine learning algorithms for practical problem-solving in data science.**
- **Utilize machine learning techniques for real-world data analysis, encompassing preprocessing, model selection, and neural network construction for classification and sequence tasks.**

Paper Title: Software Project Management

Paper Code : MDS - 2417

Course Outcomes : After successful completion of the course, students will be able to:

- **Apply project management principles to the unique context of data science projects.**
- **Demonstrate proficiency in Agile methodologies and their application in data science environments.**
- **Effectively manage risks associated with data science projects.**
- **Understand the importance of data governance and ethics in project management for data science.**

Paper Title: Research Methods and Ethics in Data Science

Paper Code: MDS – 2418

Course Outcomes : After successful completion of the course, students will be able to:

- **Design and conduct research studies in Data Science using appropriate methodologies and tools.**
- **Formulate research questions and problems specific to Data Science applications.**
- **Evaluate ethical implications of data usage and propose responsible strategies.**
- **Communicate Data Science research effectively while adhering to ethical standards.**